Stochastic FRAMES: experiment specification

Nonconvex ℓ_1-constrained least squares in dimension 1024

14 October 2026 · Uniform sampling of individual data rows

Problem. With $d = m = 1024$, the problem and its four ingredients are

$$\boxed{\min_{x\in C} f(x) + g(Tx), \qquad f(x) = \|Ax - B\|_2^2 - \frac{5}{8}(u^\top x)^2,}$$

$$C = \{x \in \mathbb{R}^{1024} : \|x\|_1 \le 1\}, \qquad T = I_{1024}, \qquad g(z) = \iota_D(z) = \begin{cases} 0, & z \in D, \\ +\infty, & z \notin D, \end{cases} \quad D = \mathbb{R}^{1024}_+.$$

Thus original feasibility means $x \ge 0$ and $\sum_j x_j \le 1$. The vectors $a, u \in \mathbb{R}^{1024}$ are

$$a_j = \begin{cases} 1/32, & 1 \le j \le 64, \\ -3/128, & 65 \le j \le 128, \\ 1/128, & 129 \le j \le 1024, \end{cases} \qquad u_j = \begin{cases} (-1)^{j-1}/8, & 1 \le j \le 64, \\ 0, & 65 \le j \le 1024. \end{cases}$$

Define a Hadamard matrix recursively and set the data matrix and response to

$$\mathsf{W}_1 = [1], \qquad \mathsf{W}_{2n} = \begin{pmatrix} \mathsf{W}_n & \mathsf{W}_n \\ \mathsf{W}_n & -\mathsf{W}_n \end{pmatrix}, \qquad \boxed{A = \frac{\mathsf{W}_{1024}\,\mathrm{diag}(c)}{\sqrt{2048}}, \qquad B = Aa.}$$

The signs $c_j \in \{-1, 1\}$ are drawn independently and uniformly once, using NumPy's default random generator with seed 20261014, and fixed for every run. Since $A^\top A = I/2$, the least-squares term is $\|x - a\|_2^2/2$. Writing $Q = I - \frac{5}{4}uu^\top$, we have

$$\nabla f(x) = Qx - a, \qquad \nabla^2 f = Q, \qquad \mathrm{spec}(Q) = \{-1/4, 1\}.$$

The shared concave quadratic makes f nonconvex; its gradient is globally 1-Lipschitz.

Row-sampled stochastic gradient. Let A_i be row i of A, and B_i the corresponding response. The finite sum and sampled gradient are

$$f = \frac{1}{m}\sum_{i=1}^m f_i, \qquad f_i(x) = m(A_i x - B_i)^2 - \frac{5}{8}(u^\top x)^2,$$

$$\boxed{q_k = 2m(A_{i_k}x_k - B_{i_k})A_{i_k}^\top - \frac{5}{4}(u^\top x_k)u, \qquad i_k \sim \mathrm{Uniform}\{1, \dots, m\}.}$$

Rows are sampled independently with replacement; the concave term is evaluated exactly. If $\mathcal{F}_k$ denotes the history before the fresh row draw, then

$$\mathbb{E}[q_k \mid \mathcal{F}_k] = \nabla f(x_k), \qquad \mathbb{E}[\|q_k - \nabla f(x_k)\|_2^2 \mid \mathcal{F}_k] = 1023\,\|x_k - a\|_2^2 \le \frac{318153}{256}.$$

Initialization and parameters. For each seed $s \in \{0, \dots, 7\}$, draw J_s uniformly from $\{1, \dots, 1024\}$ independently of all row draws, and set

$$\boxed{x_0^{(s)} = -e_{J_s}.}$$

Here e_j has entry 1 in position j and zeros elsewhere. Thus the initial point has one random entry equal to -1, with $\left\|x_0^{(s)}\right\|_1 = 1$ and $\mathrm{dist}_D(x_0^{(s)}) = 1$. Initialize momentum with the first sampled gradient. Use $N = 131072$ updates per seed, 8 seeds, float64 arithmetic, and

$$\boxed{\gamma_k = (k+1)^{-3/4}, \qquad \beta_k = (k+1)^{-1/4}, \qquad \alpha_k = (k+1)^{-1/2}, \qquad 0 \le k < N.}$$

These are respectively the Frank–Wolfe step size, smoothing parameter, and weight on the new stochastic gradient. All prefactors are 1; one row is sampled per update.

Gap and feasibility notation. Define $(x_k)_- = \min(x_k, 0)$ coordinatewise, and set

$$z_k := \nabla f(x_k) = Qx_k - a, \qquad h_k := z_k + \frac{(x_k)_-}{\beta_k}.$$

The reported gaps use these exact vectors, rather than the sampled gradient:

$$G_k := \max_{v \in C} \langle h_k, x_k - v \rangle = \boxed{\langle h_k, x_k \rangle + \|h_k\|_\infty}, \qquad \text{smoothed Frank–Wolfe gap,}$$

$$H_k := \max_{v \in C \cap D} \langle z_k, x_k - v \rangle = \boxed{\langle z_k, x_k \rangle + \max\{0, -\min_j z_{k,j}\}}, \qquad \text{original constrained gap,}$$

$$\delta_k := \operatorname{dist}_D(x_k) = \boxed{\|(x_k)_-\|_2}, \qquad \text{feasibility violation.}$$

For this quadratic objective, the gaps are explicitly

$$\boxed{\begin{aligned} G_k &= \|x_k\|_2^2 - a^\top x_k - \frac{5}{4}(u^\top x_k)^2 + \frac{\|(x_k)_-\|_2^2}{\beta_k} \\ &\quad + \max_{1 \le j \le 1024} \left| x_{k,j} - a_j - \frac{5}{4}(u^\top x_k) u_j + \frac{\min\{x_{k,j}, 0\}}{\beta_k} \right|, \\ H_k &= \|x_k\|_2^2 - a^\top x_k - \frac{5}{4}(u^\top x_k)^2 \\ &\quad + \max\left\{0, \max_{1 \le j \le 1024} \left[a_j - x_{k,j} + \frac{5}{4}(u^\top x_k) u_j \right] \right\}. \end{aligned}}$$

Since $x_k \in C$, $G_k \ge 0$. However, H_k can be negative when $x_k \notin D$; the cumulative original-gap plot therefore reports $|H_k|$. Feasibility and the original gap are interpreted together.

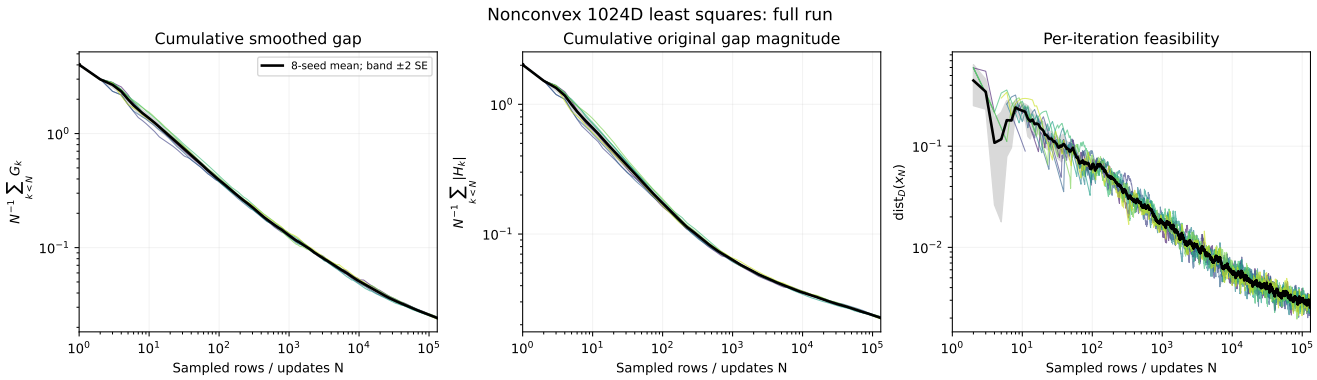
What the plots average. For $Z_k \in \{G_k, |H_k|\}$, define

$$\overline{Z}_N^{(s)} = \frac{1}{N} \sum_{k=0}^{N-1} Z_k^{(s)}, \qquad \widehat{A}_Z(N) = \frac{1}{8} \sum_{s=0}^{7} \overline{Z}_N^{(s)}, \qquad \widehat{D}(N) = \frac{1}{8} \sum_{s=0}^{7} \delta_N^{(s)}.$$

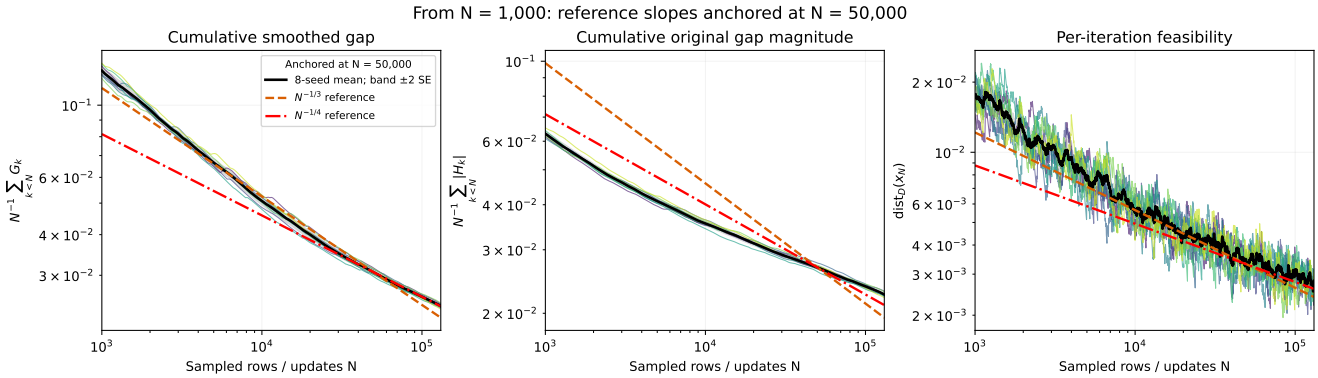
The gap averages include every pre-update iterate $x_0, \ldots, x_{N-1}$. Feasibility uses the violation at x_N after N updates, with no averaging over iterations. Thin curves show each seed's cumulative gap or current feasibility; black curves average across the eight seeds, with descriptive bands of ± 2 standard errors. The feasibility trace is empirical at every iteration: Proposition 2.11 bounds $\mathbb{E}[\delta_{k_n}]$ for $k_n \in \operatorname{argmin}_{\lceil n/2 \rceil \le k \le n} \mathbb{E}[G_k]$, rather than every last iterate.

The full-range plots have no reference lines. The zoom starts at $N = 10^3$ and retains the same cumulative gap histories, starting at $k = 0$. Its $N^{-1/3}$ and $N^{-1/4}$ reference lines both meet each panel's black curve at $N = 5 \times 10^4$; the $N^{-1/4}$ line is bright red. They illustrate slopes, not numerical upper bounds or claimed empirical rates. Long-range curves use a fixed logarithmic display grid; gap averages use all iterates, and feasibility is not smoothed over time. Zero values are omitted on logarithmic axes (all seeds have $\delta_1 = 0$).

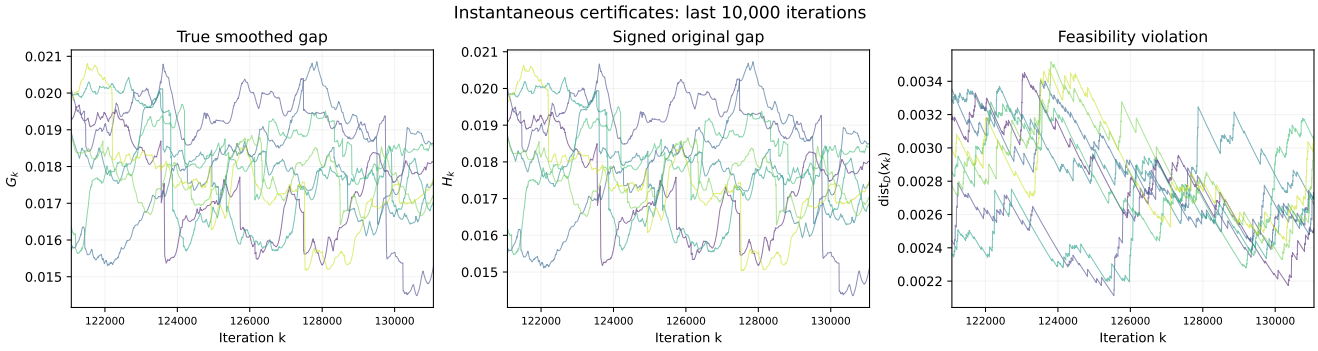
Plots. Eight seeds with random negative-vertex initialization.



Full range: cumulative averages of G_k and $|H_k|$, and current feasibility δ_N after N updates; no reference lines.



The same quantities for $N \geq 10^3$, with both reference slopes anchored at $N = 5 \times 10^4$.



Individual values of G_k, signed H_k, and δ_k during the final 10000 iterations. Each color represents one seed.